Chapter 4

Random Errors in Chemical Analysis

***Fundamentals of Analytical Chemistry*, Skoog, et al., 10th Ed., 2022**

Chapter 4 Problems:

4-3, 4-7, 4-8, 4-10

Statistical Treatment of Random Errors

Satistical Analysis only reveals information that is already present in a data set (*no new information is created*). Statistical methods allow us to characterize data and make decisions about data quality and interpretation.

A **statistical** sample does not equal the **analytical** sample.

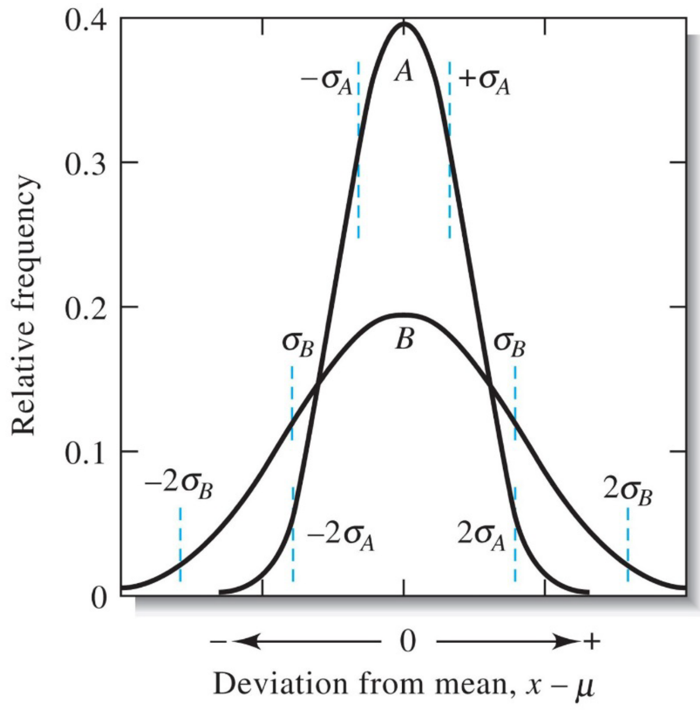
Population	**Sample**
N $\geq$ 20	**N<20**
μ = population mean	$\bar{x}$ = sample mean
In the absence of systematic error it is the accepted value	The $\bar{x}$ is an estimate of the population mean
σ = population std. dev.	s = sample std. dev.
σ is a measure of precision	s is an unbiased estimate of the population std. dev.
$\boldsymbol{\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}}$	$\boldsymbol{s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N-1}}}$

N-1 = degrees of freedom

<u>**Std. dev. has the same units as the mean**</u>

Random or indeterminate errors cannot be eliminated and are therefore present in every measurement and cause replicate analysis to fluctuate around the mean.

Gaussian distribution around the mean



(a)

Statistical Treatment of Random Errors

Regardless of width

68.3% of the area beneath a Gaussian curve for a population lies within +/- 1 σ

95.4 % lies within +/- 2 σ

99.7 % lies within +/- 3

Example 4-1

1. The following results were obtained in the replicate determination of the lead content of a blood sample: 0.752, 0.756, 0.752, 0.751, and 0.760 ppm Pb. Find the mean and the standard deviation.

Measures of Precision

Variance = s^2

Relative Standard Deviation (rsd)

$$rsd = s_r = \frac{s}{\bar{x}}$$

$$rsd_{ppt} = \frac{s}{\bar{x}} \; x \; 1000 \; ppt$$

Coefficient of Variation (CV)

The relative standard deviation multiplied by 100%

$$CV = rsd \; in \; percent = \frac{s}{\bar{x}} \; x \; 100\%$$

Spread or Range (*w*)

The difference between the largest and smallest values

For the set of data from example 4-1 calculate

a) **The variance**
b) **rsd in ppt**
c) **The coefficient of variation**
d) **The spread**

$\bar{x}$ **= 0.754 ppm Pb**

s = 0.00377 = 0.004 ppm Pb

a)

b)

c)

d)

Reporting Computed Data

Significant figures

The significant figures in a number are all the certain digits plus the first uncertain digit.

1. Disregard all initial zeros
2. Disregard all final zeros unless they follow a decimal point
3. All remaining digits including zeros between nonzero digits are significant

Significant Figures in Numerical Computations

Sums and Differences

The result should contain the same number of decimal places as the number with the smallest number of decimal places.

When adding and subtracting numbers in scientific notation express the numbers to the same power of ten.

Products and Quotients

The answer should be rounded so that it contains the same number of significant digits as the original number with the smallest number of significant digits.

Logarithms and Antilogarithms

1. In a logarithm of a number keep as many digits to the right of the decimal point as there are significant figures in the original number.
2. In an antilogarithm of a number, keep as many digits as there are digits to the right of the decimal point in the original number.

Rounding Data

When rounding a 5 always round to the nearest even number.

Propagation of Error

Standard Deviation of Calculated Results

Standard Deviation of a Sum or Difference

y = a + b –c

Standard Deviation of y = s_y = absolute std. dev. of the calculated result

$$s_y = \sqrt{{s_a}^2 + {s_b}^2 + {s_c}^2 + \cdots {s_x}^2}$$

Calculate the standard deviation of the result of

0.50 (±0.02) + 4.10 (±0.03) – 1.97 (±0.05) = ????? (± ?????)

Standard Deviation of a Product or Quotient

y = (a x b)/c

Standard Deviation of y

$$\frac{s_y}{y} = \sqrt{\left(\frac{s_a}{a}\right)^2 + \left(\frac{s_b}{b}\right)^2 + \left(\frac{s_c}{c}\right)^2 + \cdots \left(\frac{s_x}{x}\right)^2}$$

Calculate the standard deviation of the result of

<u>4.10 (±0.02) * 0.0050 (±0.0001)</u> = ????? (± ?????)

1.97 (±0.04)

Example 4-4

Calculate the standard deviation of the result of

$$\frac{[14.3 (\pm 0.2) - 11.6 (\pm 0.2)] * 0.050 (\pm 0.001)}{[820 (\pm 10) + 1030 (\pm 5)] * 42.3 (\pm 0.4)} = ????? (\pm ?????)$$